Utica University | Department of Computer Science | Fall 2026
CSC 618: Information Retrieval | Week 10 Assignment

Submission 3: Part C, Questions on the Real-Dataset Experiment

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I compared TF-IDF, BM25 and a Dirichlet Language Model in Apache Lucene on the Cranfield collection (1400 documents, 225 queries). Full methods and code are in Submission 4.

Table 1: Key results (mean over 225 queries, default parameters).

Model	MAP	MRR	P@3	P@10	nDCG@10	Recall (full list)
TF-IDF	0.2931	0.5327	0.3452	0.2244	0.3690	0.9513
BM25	0.2952	0.5204	0.3570	0.2284	0.3741	0.9523
LM-Dirichlet (μ=2000)	0.2341	0.4557	0.2726	0.1840	0.3041	0.9515

1. Did the controlled-dataset code work on the real dataset? What changed?

Yes. The metric functions in ir_metrics.py ran without modification, and an independent library (ranx) gave identical values to six decimals. The changes were all around those functions:

Change	Why
Loaders for qrels and runs	Part B data was typed into Python; Part C needs TREC-format files
Renumbering query IDs	Query numbers in the file are 1, 2, 4, 8, . . . up to 365, but qrels use 1–225 by position. Without the fix, BM25 MAP falls from 0.2952 to 0.0090, with no error raised. This was the most important change
Added cut-offs	Precision, Recall and F1 on a list of up to 1000 are nearly meaningless (Precision 0.0081), so I added them at rank 10
nDCG for all queries with binary gain	Part B had grades for Q1 only; the copy of the qrels I could obtain is binary

2. Did the real-world results match my expectations?

Partly. I expected BM25 to clearly beat TF-IDF and the Language Model to be close to BM25.

- BM25 vs TF-IDF: smaller than expected. BM25 has the higher MAP by only 0.0021, which is not significant (randomization p = 0.7492). TF-IDF has a slightly higher MRR and is the outright best model by AP on more queries (99 vs 75). Lucene's TF-IDF includes length normalisation, which probably removes much of the difference that a basic TF-IDF would show.
- LM-Dirichlet: worse than expected. Its MAP is 0.0611 below BM25 ($p < 10^{-5}$). The default μ=2000 is far above the mean document length of 102 terms. Lower μ helped (MAP 0.2757 at μ=50) but did not reach BM25. Lucene's implementation also scores only documents containing a query term and clamps negative scores to zero, so it is not a pure query-likelihood model. I did not test separately how much that contributes.
- Absolute scores: as expected for a hard small collection. A MAP near 0.30 and P@10 near 0.23 were reasonable: queries average 7.2 relevant documents, so P@10 cannot be high for many queries.

3. Which metrics were most informative, and why?

Metric	Value in this experiment
MAP	Most informative. Uses every relevant document's position, and clearly separated the Language Model
nDCG@10	Same conclusion as MAP, focused on the first page. Less informative than usual, as gains were binary
MRR	Added a distinct view: TF-IDF and BM25 swap order on it, though not significantly
P@10, Recall@10	Easy to interpret, but limited by the number of relevant documents per query (range 1–39)
Full-list P, R, F1	Least informative. Nearly identical across models (Recall 0.951–0.952), because these lists include almost all documents matching any query term

The per-query paired tests were as useful as the metrics themselves: without them, the BM25 and TF-IDF means could have been taken as a real difference.

4. What are the main limitations of my evaluation?

1. Old, narrow collection. 1960s aeronautics abstracts with verbose questions (about 18 words) do not represent modern web or enterprise search.
2. Binary relevance. The original Cranfield grades were not in the copy I could access, so nDCG did not use graded relevance here. Rows labelled "no interest" in the source were treated as non-relevant.
3. Incomplete judgments. Unjudged documents count as non-relevant, which can understate any model.
4. Parameter choice. The main comparison uses default parameters. The sensitivity results were chosen after viewing test queries, with no separate tuning set, so they are not clean performance estimates.
5. One pipeline. Results depend on Lucene's implementations and on EnglishAnalyzer, whose small stop list keeps words such as what.
6. No user data. These offline metrics say nothing about speed, satisfaction or real click behaviour.